Hanlong Li

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RESEARCH INTERESTS

Robot manipulation learning · Reinforcement & imitation learning · VLA models · Embodiment-aware control

EDUCATION

Institute of Science Tokyo — Tokyo, Japan
M.Eng. in Systems and Control Engineering — *2025 – 2027 (expected)*
Advisor: Prof. Kazuhiro Nakadai

Georgia Institute of Technology — Atlanta, GA, USA
B.S. in Computer Science — *2019 – 2024*
Threads: Intelligence & People · Graduated with High Honor

PUBLICATIONS & PREPRINTS

[1] **H. Li**, S. Tan, T. Ashizawa, B. Yen, K. Nakadai. "Paraphrase Consensus Is Not Self-Consistency: Instruction Reliance and Robustness in VLA Policies." *arXiv:2610.19438*, 2026. Under review at ACL Rolling Review. [arXiv]

[2] M. Tateno, T. Ohkawa, Y.-H. Wu, **H. Li**, T. Matsushima, Y. Sato, K. Ota. "YUBI-STAG: Contact and Semantic-Rich Alignment for VLAs via Automated Video-Language Grounding." *CoRL 2026 Workshop: The UMI Arena*, 2026. [arXiv] [project]

[3] **H. Li**, S. Tan, T. Ashizawa, B. Yen, K. Nakadai. "AMBIT: Anticipatory Multimodal Body Recruitment for Bimanual Tracking on a Humanoid." *arXiv:2609.33484*, 2026. Under review at ICRA 2027. [arXiv]

[4] **H. Li**, K. Kamalahasan, J. Li, K. Nakadai, S. Kousik. "The Talking Robot: Distortion-Robust Acoustic Models for Robot–Robot Communication." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026. [arXiv] [code] [project]

[5] B. Starbuck*, **H. Li***, B. Bras. "Toward Bio-Agentic Visual Communication and Control: Implicit Semantic Manifolds and Invariant State Estimation." *ASME International Mechanical Engineering Congress and Exposition (IMECE)*, 2026 (accepted).

[6] N. Khan, B. Wu, R. Shi, **H. Li**, B. Yen, C. T. Ishi, K. Nakadai. "Spectral Optimal Transport for Dynamics-Aware Sign Language Generation." *European Conference on Computer Vision (ECCV), MoCha Workshop*, 2026.

[7] B. Starbuck†, **H. Li**†, B. Cochran, M. Weissburg, B. Bras. "A Concept for Bio-Agentic Visual Communication: Bridging Swarm Intelligence with Biological Analogues." *Biomimetics*, 10(9):605, 2025. [DOI]

Patent application. B. Bras, M. Weissburg, B. Cochran, **H. Li**, B. Starbuck. "Biologically Inspired Active Visual Communication System and Method." U.S. Patent Application Publication US 2026/0127397 A1, 2026. [link]

*Equal contribution. †Corresponding author.

RESEARCH EXPERIENCE

Research Engineer Intern – AIRoA — Tokyo, Japan
Mentors: Nicola Scianca, Yueh-Hua (Kris) Wu — *Aug 2026 – Present*

- Initiated and lead an RL project on low-level execution of VLA action chunks: a MuJoCo/MJLab OpenArm benchmark with PPO, domain randomization, imperfect command streams and matched PD baselines. Preliminary comparison: ∼2.4% vs. ∼4.6% tracking error (RL vs. PD), a ∼48% relative decrease (pending experimental verification).

- Designed and validated on physical robots a real-time VLA execution stack: asynchronous, time-aligned action-chunk re-inference with interpolated high-frequency control and a streaming-protocol client.
- Led 60 real-robot VLA evaluation sessions (177 recorded trial segments) on a dual-arm Franka platform: experimental protocols, metrics, trajectory recording/analysis tools, and analysis of policy behavior.
- Designed and hardware-tested VLA action-representation adaptation (quaternion and rotation-vector poses, observation-relative gripper actions); co-developed hierarchical Goal/Task interfaces and a short-horizon evaluation workflow.
- Co-authored YUBI-STAG, automated video-language grounding that aligns pretrained VLAs with fine-grained manipulation language (CoRL 2026 Workshop).

Graduate Researcher – Nakadai Lab Institute of Science Tokyo
Advisor: Prof. Kazuhiro Nakadai *Oct 2025 – Present*

- Led AMBIT, a CVAE-based framework for anticipatory whole-body recruitment in humanoid bimanual tracking (CVAE strategy generation, a non-learned verifier, and a whole-body differential-IK QP tracker): 85% vs. 74% success over a reactive tracker on 160 held-out MuJoCo episodes, 0.85 vs. 0.53 on a Unitree G1, and hardware execution on an externally supported Unitree R1.
- Led a study of instruction reliance in VLA policies: rewording LIBERO instructions costs base OpenVLA 38.0 points, paraphrase consensus does not recover it, and paraphrase LoRA post-training reduces the effect to +2.0 vs. −32.0 points for a matched control.
- Co-authored an ECCV 2026 workshop paper on sign language generation with spectral optimal transport.

Visiting Researcher – Robot Danger Lab Georgia Tech
Advisor: Prof. Shreyas Kousik *Jun 2024 – Sep 2025*

- Developed a learned acoustic communication system for robot–robot messaging (first-author IROS 2026 paper): a FastSpeech-style transmitter and Conformer-CTC receiver co-trained as a lightweight codec for 128-token robot command sequences.
- Designed a procedural-synthesizer warm start and channel-augmentation curriculum to stabilize end-to-end training under noise and distortion.
- Achieved 8.3% character error rate at 0 dB SNR; 2.1M parameters, under 13 ms on CPU.

Undergraduate Researcher, then Part-time Researcher – SDM Lab Georgia Tech
Advisor: Prof. Bert Bras *Sep 2023 – Aug 2025*

- Co-developed bio-agentic visual communication for RF-degraded UAV swarms: a GenAI/RAG pipeline mapping perceived visual signals to symbolic responses (trajectories, LED patterns).
- Developed a bio-inspired AprilTag signaling system with ROS perception and control components; improved tag detection range fivefold with a YOLO model, reaching 96% accuracy across lighting conditions.
- Evaluated an ISM-IEKF benchmark: mean execution error 10.48 m → 6.37 m over 200 interception trials (IMECE 2026).

INDUSTRY EXPERIENCE

Before moving into full-time robotics research, I worked as a full-time software engineer (2024–2025), building full-stack features for AI-powered administrator tools and integrating machine-learning outputs into production web and iOS applications.

TEACHING & HONORS

Teaching Assistant, Global PBL Foundation, Institute of Science Tokyo (2025) · Dean's List, Georgia Tech

SKILLS

Robot learning & control: PyTorch, JAX, PPO/RL, CVAE, QP/IK, ROS 2, MuJoCo/MJLab, Isaac Sim, Gazebo
Programming & systems: Python, C/C++, Java, JavaScript/TypeScript, Linux, Docker, Git, Jetson, OpenCV
Languages: English (bilingual), Mandarin (bilingual), Japanese (basic)